

LONDRINA, Brazil

WMO: 837680

Lat: 23.334S

Lon: 51.130W

Elev: 569

StdP: 94.67

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	8.1	10.0	1.7	4.6	16.0	4.0	5.4	18.0	6.7	21.0	6.0	20.7	2.2	220	0.290

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.8	34.2	21.6	33.2	21.9	32.1	22.0	25.6	29.2	25.1	28.8	24.7	28.3	3.6	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.0	21.5	27.1	24.2	20.4	26.4	23.8	20.1	26.2	82.7	29.5	80.3	28.5	78.3	28.6	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.8	5.9	5.2	4.9	36.4	2.2	1.3	3.3	37.3	2.0	38.0	0.8	38.7	-0.8	39.7		
(5)			3.2	26.8	2.3	1.0	1.5	27.5	0.2	28.1	-1.1	28.6	-2.8	29.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.7	25.2	25.4	25.1	23.3	19.8	18.6	18.7	20.6	22.1	23.7	24.3	25.2
	DBStd	3.62	1.70	1.81	1.87	2.55	2.78	2.99	3.08	3.24	3.49	2.89	2.20	1.99
	HDD10.0	1	0	0	0	0	0	0	1	0	0	0	0	0
	HDD18.3	109	0	0	0	2	17	32	33	17	7	1	0	0
	CDD10.0	4620	471	432	467	399	305	259	270	328	364	425	428	471
	CDD18.3	1686	213	199	208	151	64	41	44	87	122	167	178	213
	CDH23.3	13235	1544	1503	1586	1111	361	217	306	768	1165	1457	1468	1750
CDH26.7	4635	511	537	564	355	67	22	51	245	496	601	543	645	
Wind	WSAvg	2.3	2.4	2.2	2.2	2.1	2.0	1.9	2.0	2.3	2.6	2.7	2.8	2.5
Precipitation	PrecAvg	1532	230	193	126	92	118	86	67	46	121	129	151	173
	PrecMax	2134	426	366	308	218	325	355	237	183	334	334	375	306
	PrecMin	1201	47	61	21	0	16	1	0	0	15	4	51	12
	PrecStd	236	94	79	66	50	78	79	62	46	70	84	73	81
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.0	35.1	34.0	33.1	30.2	28.8	30.0	32.9	35.9	36.2	34.9	34.2
		MCWB	23.6	21.7	23.3	22.2	20.6	20.3	18.3	18.6	20.1	21.5	22.0	22.5
	2%	DB	32.0	32.8	32.2	31.8	28.2	27.1	28.1	31.0	33.9	34.1	32.9	32.9
		MCWB	23.7	23.1	23.1	22.1	20.8	20.0	18.3	18.4	19.3	21.8	22.1	22.8
	5%	DB	30.9	31.1	31.1	30.1	27.0	25.9	26.7	29.2	31.9	32.1	31.2	31.8
		MCWB	23.6	23.6	22.9	21.7	20.2	19.5	18.3	18.2	19.2	21.5	21.5	22.9
	10%	DB	29.5	30.0	30.0	28.9	25.2	24.2	25.0	27.9	29.8	30.0	29.9	30.1
		MCWB	23.4	23.5	22.6	21.5	19.5	19.0	18.0	18.9	18.9	21.1	21.5	22.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.3	26.3	26.2	24.9	23.0	22.6	21.6	22.3	23.4	24.7	25.3	25.8
		MCDB	29.2	29.2	29.6	28.4	27.4	25.5	25.3	26.7	29.2	29.6	29.7	29.9
	2%	WB	25.4	25.5	25.1	23.9	22.1	21.2	20.5	20.7	22.2	23.8	24.5	25.1
		MCDB	28.7	28.7	28.9	27.8	25.9	24.7	24.5	25.8	27.8	29.4	28.7	29.3
	5%	WB	24.9	24.9	24.5	23.2	21.2	20.4	19.7	19.8	21.4	23.1	23.7	24.5
		MCDB	28.3	28.1	28.2	27.4	25.0	24.0	23.9	25.6	26.7	28.4	27.8	28.5
	10%	WB	24.4	24.3	23.9	22.6	20.3	19.7	18.8	18.9	20.5	22.4	23.1	23.9
		MCDB	27.6	27.6	27.5	26.7	24.0	23.2	22.8	25.0	25.7	27.5	26.9	27.6
Mean Daily Temperature Range	5% DB	MDBR	8.5	8.8	9.6	10.1	9.5	9.7	10.9	12.3	11.8	11.0	10.6	9.4
		MCDBR	10.5	10.8	11.2	11.4	11.5	11.5	13.1	14.4	14.9	14.0	12.7	11.6
		MCWBR	4.3	4.0	4.2	4.2	5.1	5.2	5.5	5.2	4.9	4.8	4.6	4.4
	5% WB	MCDBR	8.5	8.7	9.5	9.6	9.7	9.7	10.8	12.1	11.9	11.0	9.7	9.1
		MCWBR	4.3	4.3	4.3	4.3	4.9	5.2	5.7	5.4	5.5	4.9	4.6	4.4
Clear-Sky Solar Irradiance	taub	0.394	0.394	0.387	0.367	0.352	0.361	0.364	0.423	0.528	0.457	0.410	0.401	
	taud	2.432	2.426	2.442	2.486	2.486	2.452	2.427	2.219	1.933	2.205	2.348	2.394	
	Ebn at Noon	950	936	916	892	863	831	841	820	774	868	929	945	
	Edh at Noon	124	122	115	101	93	92	98	130	188	151	134	129	
All-Sky Solar Radiation	RadAvg	5.82	5.74	5.40	4.80	3.77	3.44	3.72	4.54	5.00	5.45	6.01	6.12	
	RadStd	0.59	0.45	0.40	0.34	0.38	0.34	0.39	0.32	0.49	0.46	0.47	0.56	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page